

Exercise to laboratory 4: The Munk's theory

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April 24, 2026

Deadline: Monday June 1, 2026.

1 Theory (10P)

- i. State the necessary assumption that are needed to study the depth integrated gyre circulation. (2.5P)
- ii. Write down the definition of the barotropic stream function. (1P)
- iii. Write down the equation for the stream function when assuming boundary condition $\Psi_s(x_e) = 0$. State the name of the circulation that is described by the equation. (1P)
- iv. Describe the shape of the gyre. (1P)
- v. State the additional assumptions for the Munk theory (1.5P)
- vi. Describe the main changes in the shape of the stream function under Munk's theory. (1P)
- vii. Give the equation for the Munk solution. (1P)
- viii. Describe the (shape of the) wind stress (discuss the equation for the wind stress) and the geographical situation where this can commonly be applied. (1P)

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2 Experiments

Set up the model as described in the laboratory description for laboratory 4
- Munk's theory and follow the instruction for the experiments.

2.1 Sverdrup circulation (10P)

- i. Provide plots of the stream function for all your experiments at appropriate time steps. (4P)
- ii. What effect does $\beta = \partial f / \partial y$ have on the system? (1P)
- iii. What effect does the wind forcing have? (1P)
- iv. Provide cross sections (in x-direction) of the stream functions computed from your experiments. (1P)
- v. From all you plots, discuss if the results resemble theory? Why / Why not? (1P)
- vi. Do the results improve if the size of the domain is increased? (1P)
- vii. What are the limitations of the Sverdrup's model? (1P)

2.2 Munk's theory (10P)

- i. Provide plots of the stream functions of your experiments. (4P)
- ii. Describe how the results have changed from Sverdrups circulation to Munk's theory. (1P)
- iii. Do theory and numerical solution correlate to each other? (1P)
- iv. Provide
 - cross sections (in x-direction) of the stream functions computed from your experiments. (1P)
 - the difference of the cross section of stream functions between theory and computed from simulation results. (1P)and make sure that they also show the Munk thickness from theory. (1P)
- v. From that, discuss the role of the friction coefficient for the shape. (1P)

Good luck!